Changliang Zhou

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Profile

I am a Ph.D. student at the Southern University of Science and Technology, specializing in AI for Optimization. My research focuses on deep reinforcement learning for complex combinatorial optimization problems (COPs), such as vehicle routing problems (VRPs), with an emphasis on tackling key challenges like **out-of-distribution zero-shot generalization** and **domain foundation model construction**. Furthermore, I lead the development of EasyCO, a standardized, highly scalable, general-purpose platform for combinatorial optimization. It currently unifies workflows for 50+ solvers and supports automated solving for 150+ COPs.

Education

Linyi University, B.Eng. degree in Software Engineering, School of Information Science and Engineering — Sep. 2014–Jun. 2018

Qingdao University, M.Eng. degree in Software Engineering, College of Computer Science and Technology — Sep. 2019–Jun. 2022

Southern University of Science and Technology, Ph.D. student in Intelligent Manufacturing and Robotics, School of Automation and Intelligent Manufacturing — Sep. 2022–Present

Highlights

ICAM: Instance-Conditioned Adaptation Model for Thousand-Scale Generalization [IEEE T-ITS 2026] Code & Paper — 2023–2024

- Introduced a lightweight instance-conditioned adaptation function and a low-complexity attention mechanism to adjust policies using instance-specific information.
- Validated the method on four representative routing problems (TSP, CVRP, ATSP & CVRPTW), improving generalization while preserving **low inference time (at the second level)**.

L2R: Learning to Reduce Search Space for Million-Scale Generalization [KDD 2026] Code & Paper — 2024–2025

- Proposed the first learning-based framework to dynamically prune the search space by adaptively prioritizing nodes based on problem-specific features and states.
- Developed **the first** specialist neural model for **10-million-node** VRP instances. Trained only on **100-node** instances, it generalizes to TSP10M in a zero-shot manner with a 5.05% gap.

URS: A Powerful Domain Foundation Model for Cross-Problem Zero-Shot Generalization [ICML 2026] Code & Paper — 2025–2026

- Proposed a unified data representation that eliminates the need for problem enumeration, significantly broadening problem coverage by projecting diverse VRP variants into a unified feature space.
- Developed **the first** domain foundation model capable of solving **100+** VRP variants with a single model without retraining or fine-tuning, while successfully scaling to **7,000**-node instances.

EasyCO: A Learning-Driven Platform for Combinatorial Optimization [In Development] (Lead) — 2025–Present

- Established a standardized pipeline spanning data generation, environment setup, training, and evaluation.
- Currently supports standardized workflows for **50+** solvers and automated solving for **150+** combinatorial optimization problems.

Publications

As of June 30, 2026, I have published **11 papers**, including **4 first/co-first-author papers**. My work has appeared in venues including **ICML, KDD, and IEEE T-ITS**, and has received **258 citations** on Google Scholar. Selected publications are listed below:

- **Changliang Zhou**, Canhong Yu, Shunyu Yao, Xi Lin, Zhenkun Wang†, Yu Zhou, Qingfu Zhang. URS: A Unified Neural Routing Solver for Cross-Problem Zero-Shot Generalization. *43rd International Conference on Machine Learning* (**ICML**), 2026.
- **Changliang Zhou***, Xi Lin*, Zhenkun Wang†, Qingfu Zhang. Learning to Reduce Search Space for Generalizable Neural Routing Solver. *32nd SIGKDD Conference on Knowledge Discovery and Data Mining* (**KDD**), 2026.
- **Changliang Zhou***, Xi Lin*, Zhenkun Wang†, Xialiang Tong, Mingxuan Yuan, Qingfu Zhang. Instance-Conditioned Adaptation for Large-Scale Generalization of Neural Routing Solver. *IEEE Transactions on Intelligent Transportation Systems* (**T-ITS**), 2026.
- Zhi Zheng, **Changliang Zhou**, Xialiang Tong, Mingxuan Yuan, Zhenkun Wang†. UDC: A Unified Neural Divide-and-Conquer Framework for Large-Scale Combinatorial Optimization Problems. *Advances in Neural Information Processing Systems* (**NeurIPS**), 2024.
- Yubin Xiao, Di Wang, Boyang Li, Mingzhao Wang, Xuan Wu, **Changliang Zhou**, You Zhou†. Distilling Autoregressive Models to Obtain High-Performance Non-autoregressive Solvers for Vehicle Routing Problems with Faster Inference Speed. *Proceedings of the AAAI Conference on Artificial Intelligence* (**AAAI**), 2024.
- Canhong Yu, **Changliang Zhou**, Rongsheng Chen, Zhenkun Wang†, Yu Zhou. Rethinking Constraint Awareness for Efficient State Embedding of Neural Routing Solver. *arXiv preprint arXiv:2605.10122*, 2026.
- Rongsheng Chen, **Changliang Zhou**, Canhong Yu, Yuanyao Chen, Yu Zhou, Zhuo Chen, Zhenkun Wang†. SPACE: Unifying Symmetric and Asymmetric Routing Problems for Generalist Neural Solver. *arXiv preprint arXiv:2605.24484*, 2026.
- Yunpeng Ba, Xi Lin, **Changliang Zhou**, Ruihao Zheng, Zhenkun Wang†, Xinyan Liang, Zhichao Lu, Jianyong Sun, Yuhua Qian, Qingfu Zhang. Survey on Neural Routing Solvers. *arXiv preprint arXiv:2602.21761*, 2026.

  * Equal contribution   † Corresponding author   ____ Co-mentored student

Service and Honors

- **Reviewer**: ICLR 2026; NeurIPS 2026.
- **University Honor**: Outstanding Postgraduate Student, Southern University of Science and Technology, 2022–2023.